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library(forecast)
library(fpp2)
library(urca)
library(tidyverse)
#view the data to assess stationarity:
autoplot(usmelec)
#Investigate number of differences required:
#do we need a seasonal difference?
nsdiffs(log(usmelec))
#do we need more g=l differencing?
ndiffs(diff(log(usmelec),12))
#do we need more g=l differencing?
ndiffs(diff(diff(log(usmelec),12),1))

#KPSS test:
#original data:
autoplot(usmelec)
usmelec %>% ur.kpss() %>% summary()
#Test statistic > 1% critical value
# -> Reject H0, time series is not stationary

#seasonally differenced data:
autoplot(diff(log(usmelec),12))
diff(log(usmelec),12)%>% ur.kpss() %>% summary()
#Test statistic > 1% critical value
# -> Reject H0, time series is not stationary

#differenced seasonally differenced data
#(i.e. 2nd-order differencing):
autoplot(diff(diff(log(usmelec),12),1))
diff(diff(log(usmelec),12),1)%>% ur.kpss() %>% summary()
#Test statistic < 1% critical value
# -> Do not reject H0, time series is stationary

```